Aram Ebtekar

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Employment History

2025 – 2026 **University of California, Berkeley, Research Collaborator.** Mathematics of safe reinforcement learning agents, with Michael Cohen. Other activities: Nature reviewer, AGI-25 program committee, Universal AI international reading group co-organizer, Principles of Intelligence research mentor.

2025 **Anthropic, AI Safety Research Fellow.** Co-developed the inoculation prompting technique to mitigate reward hacking in Large Language Models.

2021 – 2024 **Independent, Researcher.** Published one empirical and three theoretical papers on Bayesian filtering, the causal arrow of time, algorithmic thermodynamics, and root cause analysis. I also took on contracts deploying my research (the Elo-MMR algorithm), and optimizing autonomous vehicle planning algorithms for Caterpillar.

2018 – 2019 **Mythic, Senior AI Research Scientist.** Led two exploratory efforts: video super-resolution, covering the whole pipeline from realistic dataset collection to neural architecture design; and hardware co-design investigations on how efficiently convolutional layers map to hardware, and how long the resulting models retain their accuracy.

2016 – 2018 **Waymo, Behavior Prediction Research Engineer.** Framed driver behavior as trajectory optimization problems, enabling autonomous cars to predict surrounding drivers' movements in real-time; and led a study group to spread expertise in deep reinforcement learning techniques.

Education

2019 – 2022 **Self-directed studies.** An eclectic mix of graduate-level math, physics, economics, statistics, information theory, and undergrad-level humanities, gathered from a combination of UBC courses, MIT OpenCourseWare, textbooks, papers, and more, as preparation for technical interdisciplinary research.

2012 – 2015 **Carnegie Mellon University, M.S. (Ph.D. candidate)** in Computer Science.
- NSERC Canada Graduate Scholarship M
- Research projects in hybrid systems verification and search-based planning
- Completed the 2012 Summer School in Algorithmic Economics
- Memberships: Graduate Student Assembly departmental representatives, Ballroom Dance Club, School of Computer Science musical performances

2008 – 2012 **University of British Columbia, B.Sc. Honours** in Computer Science & Mathematics.
- Research projects in evolutionary game theory and computational geometry
- GPA: 92% (A+)

Research Publications

1. A. Ebtekar and M. K. Cohen, "Golden handcuffs make safer AI agents," *submitted to Conference on Learning Theory*, 2026.
2. N. Wichers, A. Ebtekar, A. Azarbal, *et al.*, "Inoculation prompting: Instructing LLMs to misbehave at train-time improves test-time alignment," 2026. arXiv: 2510.05024 [cs.LG].
3. A. Ebtekar and M. Hutter, "Foundations of algorithmic thermodynamics," *Physical Review E*, 2025. DOI: 10.1103/PhysRevE.111.014118.
4. A. Ebtekar, Y. Wang, and D. Janzing, "Toward universal laws of outlier propagation," *41st Conference on Uncertainty In Artificial Intelligence*, 2025. URL: https://dl.acm.org/doi/10.5555/3762387.3762440.

5 A. Ebtekar and M. Hutter, "Modeling the arrows of time with causal multibaker maps," *Entropy* ***cover article, Editor's Choice***, vol. 26, no. 9, p. 776, 2024. DOI: 10.3390/e26090776.

6 A. Ebtekar and P. Liu, "Elo-MMR: A rating system for massive multiplayer competitions," in *Proceedings of 30th The Web Conference*, 2021, pp. 1772–1784. DOI: 10.1145/3442381.3450091.

Skills

Nat. Languages	Fluent in English, French, Persian. Beginner in Spanish, Mandarin, Japanese.
Prog. Languages	Rust, C++17, LaTeX, Python, PyTorch, Keras, Java, C.
Academic	Theoretical computer science, mathematics, statistics, physics, philosophy, economics, robotics, machine learning, software design, technical writing, teaching.

Miscellaneous Experience

Contest Achievements

- 2015 61st place globally among over 50,000 registrants in the Google Code Jam.
- 57th place globally in the Topcoder Open Algorithm Competition.
- 6th place in the North American Invitational Programming Contest's Open Division, despite competing solo against teams of up to three.
- Codeforces Grandmaster, former 1st in Canada, peak rating 2400+ on Codeforces and Topcoder.
- ACM ICPC Pacific Northwest regional contest problem setter, author of problems J,L,N.
- 2012 18th place in the ACM ICPC World Finals in Warsaw, Poland.
- 2011 Top 250, Team Honorable Mention in the William Lowell Putnam Mathematical Competition.
- UBC Thunderbots team, 9th place in the RoboCup SSL international robot soccer competition.

Selected Projects

- 2020 **Technical Blogging.** My most popular article on foundations of probability made the front page of Hacker News and received over 200 comments.
- 2017 **Rust Algorithms Cookbook.** A collection of classic algorithms and data structures elegantly crafted in Rust, serving as a proof of concept of the language's compile-time safety discipline in contest programming. On 20/06/2017, it was the #1 trending GitHub repository globally.
- 2012 **U! Robot!** Lead engineer in a team of 8 developers at a 48-hour Global Game Jam, completing a platforming game that was showcased in the top 3.